

Master Architecture of the DBHE-Driven Heating, Cooling and Dehumidification Plant

Streams, Junctions, Valve Network, Operating Modes,
Load Interface, and Verification Protocol

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Abstract

This document specifies the *master architecture* of the surface plant driven by a closed-loop Deep Borehole Heat Exchanger formed from a repurposed legacy oil-and-gas well. Where the companion model report derives the governing equations of each component, this report defines the *topology*: which streams exist, where they join and split, which valve or damper controls each branch, and which subset of components is active in each operating mode. The architecture is deliberately a superset. It contains every component the plant could need across the full plane of sensible and latent loads, and a given operating mode is obtained by activating a subset of branches rather than by rebuilding the plant. Section 5 enumerates the resulting modes; Section 7 records which components are implemented in the code and which are pending.

Every numerical value in this report is extracted at build time from the model itself — the notebook, the frozen regression fixtures, and the parametric sweep archives. The document therefore cannot disagree with the code it describes. It is regenerated whenever the model version changes.

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1 Scope, and how this document stays current

1.1 What is specified here

The plant comprises five fluid circuits: the geothermal water loop, a chilled water loop, a cooling-tower water loop, the process (building supply) air stream, and the scavenging (regeneration) air stream. The master architecture is the union of all flow paths among them. It is not a single machine but a family: each operating mode activates a subset of the branches, and the inactive branches are present but idle.

Two consequences follow, and they shape the whole study. First, the number of distinct plants is finite and small once the valves and dampers are treated as two-position devices rather than continuously modulating ones. Second, the question “which plant should be built” becomes “which subset of the master architecture serves the load quadrants that actually occur, using the water temperature the well can actually deliver over the design life”. That is the question Sections 5 and 6 address.

1.2 Generation, not transcription

This report is emitted by `report/build_report.py` in the model repository. The build step reads:

- the notebook, for the version stamp and the configuration constants;
- `verification/reference/v*_baseline.json`, the frozen per-version results recorded by the regression harness;
- the parametric sweep archives, for the resource-feasibility tables.

No number below is typed by hand. When the model advances to the next version, the report is rebuilt and republished, and the version history in Section 9 gains a row automatically. A statement in this document that contradicts the code is therefore a defect in the build script, not a stale paragraph.

2 Nomenclature convention

State labels follow a two- or three-letter convention. The first subscript letter identifies the stream: w for geothermal water, c for chilled water, p for process air, r for scavenging (regeneration) air, t for cooling-tower water. Subsequent letters identify the station. Valves on water circuits are $v_A \dots v_H$; dampers on air circuits are $d_{pA} \dots d_{pI}$ for the process stream and d_{rA}, d_{rB} for the scavenging stream.

Symbol	Circuit	Station
T_{in}	water	injection into the annulus
T_{ret}	water	production from the tubing
T_{wr}	water	after the v_F well-bypass rejoins the production stream
T_{wa}	water	accumulator outlet
T_{wg}	water	absorption-generator outlet
T_{wma}	water	generator-skip branch merged with the v_B accumulator bypass
T_{wmg}	water	T_{wg} tempered by T_{wma} through v_D
T_{wx}	water	regeneration heat-exchanger outlet

Symbol	Circuit	Station
T_{wmx}	water	after the v_H remainder blends back; returns to the pump
T_{pc}, w_{pc}	process air	building comfort state (return)
T_{pmc}, w_{pmc}	process air	mixed state entering the supply fan
T_{pd}, w_{pd}	process air	desiccant-wheel process outlet
T_{pmo}, w_{pmo}	process air	after fork 1 remerges, entering fork 2
T_{pi}, w_{pi}	process air	building supply
T_{ea}, RH_{ea}	process air	building exhaust
T_{rm}, w_{rm}	scavenging air	mixed state entering the scavenging fan
T_{ri}, w_{ri}	scavenging air	regeneration-coil inlet
T_{ro}, w_{ro}	scavenging air	regeneration-coil outlet, wheel inlet
T_{rd}, w_{rd}	scavenging air	wheel regeneration outlet, vented

3 The five circuits

3.1 Geothermal water

The water circuit is a closed loop with four decision points. Reading from the pump:

1. v_E / v_F — **well bifurcation**. v_E sends flow down the annulus; v_F bypasses the well entirely and rejoins the production stream. The mixed state is T_{wr} . This decouples the loop flow from the well flow, so the surface plant can be run at a flow the subsurface would not tolerate, and vice versa.
2. v_A / v_B — **accumulator**. v_A charges or discharges the thermal store; v_B bypasses it. The accumulator is a buffer, not a topology choice: its state varies through the day within any mode.
3. v_C — **absorption generator**. The branch that skips the generator merges with the v_B bypass to form T_{wma} .
4. v_D , **then** v_G / v_H . v_D blends T_{wma} into the generator outlet T_{wg} , raising the regeneration-coil inlet to T_{wmg} . v_G passes that stream through the coil; v_H carries the remainder of T_{wma} around the coil to blend with the coil outlet T_{wx} , giving T_{wmx} .

The pairing of v_D and v_G deserves emphasis because the two are easily confused. v_D **raises** the coil inlet temperature: it can only blend *hotter* bypass water into a generator outlet that the chiller has cooled. v_G **throttles**: reducing the share of flow through the coil lowers the coil duty, hence the regeneration air temperature T_{ro} , hence the drying the wheel performs. When the wheel must be turned *down*, v_G is the control and v_D is not.

3.2 Process air

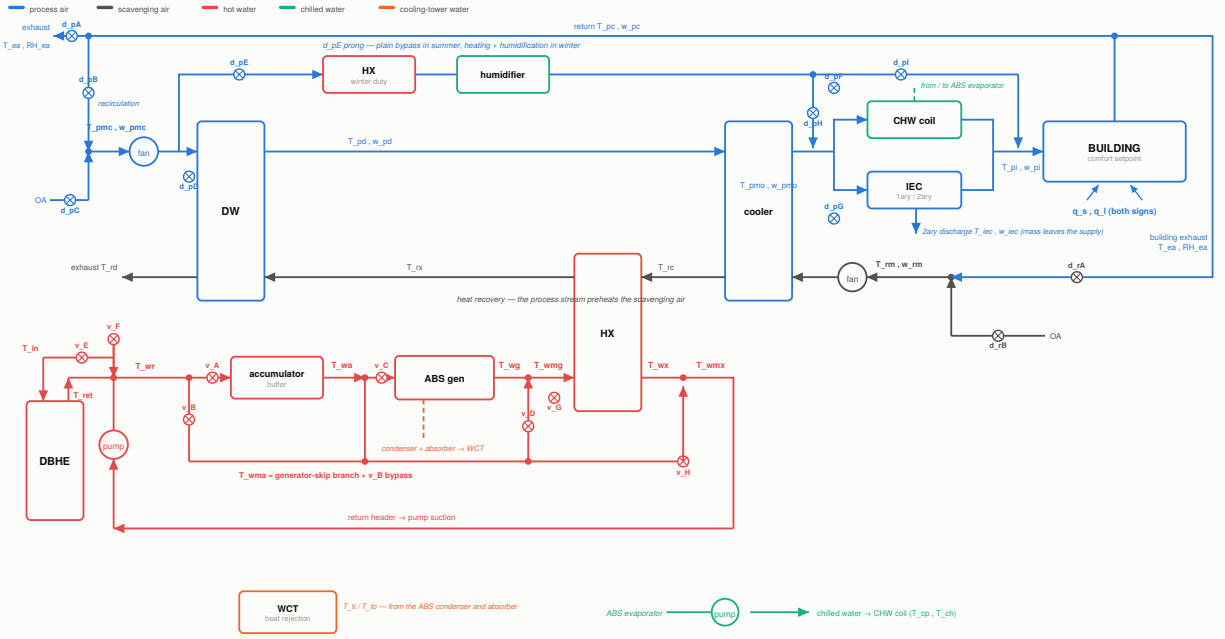
$$\text{mix} \rightarrow \text{fan} \rightarrow \underbrace{\left\{ d_{pD} \rightarrow \text{DW} \rightarrow \text{cooler} \mid d_{pE} \rightarrow \text{HX}_{\text{winter}} \rightarrow \text{humidifier} \right\}}_{\text{fork 1}} \rightarrow T_{pmo} \rightarrow \underbrace{\left\{ d_{pF} \rightarrow \text{CHW} \mid d_{pG} \rightarrow \text{IEC} \right\}}_{\text{fork 2}}$$

The building return stream bifurcates at a node: d_{pA} exhausts, d_{pB} recirculates, and the recirculated air joins outdoor air admitted through d_{pC} at a mixing junction whose state is T_{pmc}, w_{pmc} .

Fork 1 selects between conditioning (wheel followed by the recuperative cooler) and the alternative prong that carries the winter heating coil and the humidifier. The d_{pE} prong terminates in

Master architecture

Junctions rather than mixing boxes; $d_{pA/d_{pB/d_{pC}}$ and $d_{rA/d_{rB}}$ on the branches themselves



Water circuit drawn in reduced form — see the dedicated geothermal water figure for the same chain at full size. ABS sub-circuits and the chilled-water loop are shown as labelled stubs for legibility.

Figure 1: Master architecture. Every component the plant could need across the full load plane is present; an operating mode is a subset of active branches, not a different machine.

Geothermal water circuit

v_D tempers the generator outlet before the HX; v_H blends the remainder after it

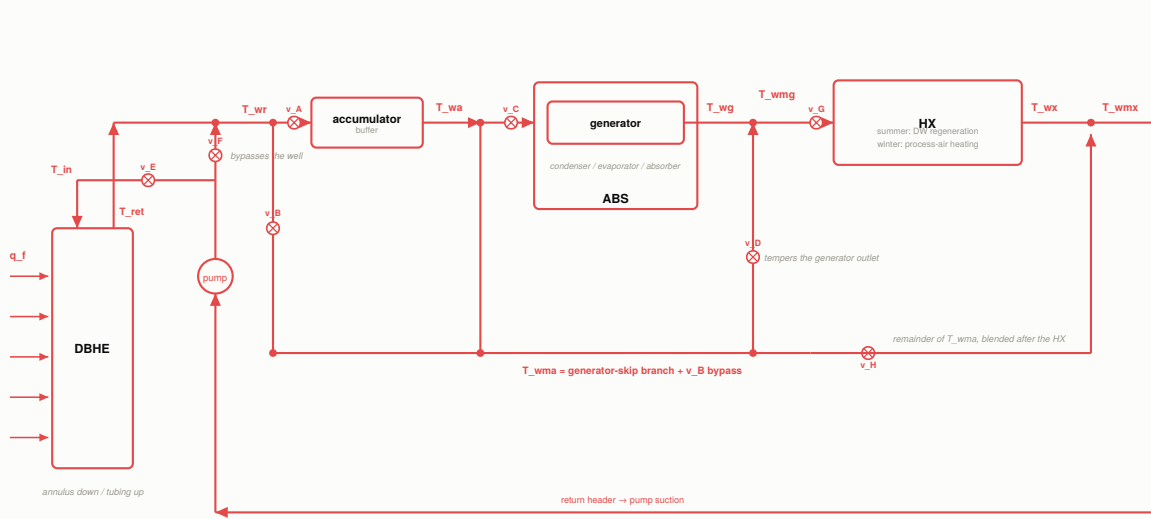


Figure 2: Geothermal water circuit. v_E/v_F decouple loop flow from well flow; v_A/v_B gate the accumulator; v_C gates the absorption generator; v_D raises the coil inlet, while v_G/v_H throttle the coil duty.

two dampers: d_{pH} returns it to T_{pmo} , upstream of fork 2, which is the summer bypass; d_{pI} carries it past fork 2 directly into the supply duct, which is the winter path. Fork 2 is a genuine parallel pair — chilled-water coil *or* indirect evaporative cooler, never both — rejoining at the supply state T_{pi}, w_{pi} .

Fork 2 needs no third “neither” prong: with no chilled water circulating and no evaporative water supplied, air passes either branch at nothing but a pressure drop.

3.3 Scavenging air

mix(d_{rA} building exhaust + d_{rB} outdoor air) $\rightarrow$ fan $\rightarrow$ cooler $\rightarrow$ HX $\rightarrow$ DW $\rightarrow$ vented

Two features of this circuit carry more weight than their simplicity suggests.

First, the scavenging stream passes through the **cooler before the regeneration coil**, so the process stream’s rejected heat preheats it and the coil has less lifting to do. The cooler is therefore a recuperator between the two air streams, not a heat rejection device.

Second, the mixing junction draws on **building exhaust and outdoor air**, and never on the wheel’s own regeneration outlet. Recirculating the wheel exhaust returns its desorbed moisture to its own inlet, which collapses the relative-humidity difference the wheel drives on. The code did exactly that until v10.5; correcting it nearly tripled moisture removal.

A physical bound constrains the mix. The building can only exhaust what it admits as fresh air, so the exhaust available to the scavenging loop is $\dot{m}_{\text{vent}} = 10.00 \text{ kg/s}$ against a scavenging demand of $\dot{m}_{\text{reg}} = 23.52 \text{ kg/s}$. The building-exhaust fraction is therefore capped at 0.425 and the balance made up with outdoor air.

3.4 Chilled water and heat rejection

The absorption chiller is a single indivisible subsystem with its chilled-water distribution: a generator fed by geothermal water, an evaporator serving the chilled-water coil in the process stream, and a condenser and absorber rejecting to a wet cooling tower. A chiller with nowhere to send chilled water is not a configuration, so **ABS and CHW are one switch, not two**.

4 The load interface

The plant sees the building only through a steady, well-mixed pair of balances. With $q_s > 0$ meaning cooling is required and $\dot{m}_w > 0$ meaning dehumidification is required,

$$q_s = \dot{m}_{\text{pro}} c_p (T_{pc} - T_{pi}), \quad (1)$$

$$\dot{m}_w = \dot{m}_{\text{pro}} (w_{pc} - w_{pi}). \quad (2)$$

Two equations, three unknowns ($\dot{m}_{\text{pro}}, T_{pi}, w_{pi}$). **Exactly one closure must be supplied**, and which one is a design decision rather than a physical fact:

- give $\dot{m}_{\text{pro}}$ — constant volume; the supply state follows;
- give $\Delta T_{\text{sup}} = T_{pc} - T_{pi}$ — variable volume; the flow follows.

Dividing (1) by (2) eliminates the flow:

$$\frac{T_{pc} - T_{pi}}{w_{pc} - w_{pi}} = \frac{q_s}{\dot{m}_w} \frac{1}{c_p}. \quad (3)$$

The load *ratio* alone fixes the direction of the room condition line on the psychrometric chart; the closure fixes how far along that line the supply state sits. This is the classical construction, and it is why the interface needs exactly one more piece of information and no more.

Latent load is carried as a moisture rate, not as a power. Expressing q_l in kilowatts requires choosing h_{fg} , and inverting (2) requires choosing it again. If the two choices differ — and values between 2450 and 2501 kJ/kg are all in common use — w_{pi} is wrong by about 2% with nothing on screen to show it. The interface therefore takes $\dot{m}_w$ in kg/s and offers a convenience wrapper that converts from kilowatts with an explicit h_{fg} .

The closure in the hatchery case is derived, not given. The supply flow is set by a design sensible load carried at a prescribed supply depression, floored by the forced ventilation requirement:

$$\dot{m}_{pro} = \max\left(\frac{Q_{s,design}}{c_p \Delta T_{sup,design}}, \dot{m}_{vent}\right) = \max(33.60 \text{ kg/s}, 10.00 \text{ kg/s}) = 33.60 \text{ kg/s}. \quad (4)$$

The depression rule binds. Its consequence reaches much further than fan sizing: a supply depression of only 3 K demands a large airflow, and a large airflow needs only a very small humidity difference to carry the same latent load. The wheel is not underperforming in this plant — it is barely being asked to work.

5 Operating modes

5.1 Naming

Load quadrants are named by the *processes they require*, from the sign pair ($\text{sgn } q_s, \text{sgn } q_l$): cooling or heating, combined with dehumidification or humidification. Seasonal names were considered and rejected: they are climate-relative, since hot-and-dry is the only summer in a semi-arid region, and setpoint-relative, since the quadrant boundaries sit at the comfort state rather than at some neutral condition.

The sensible heat ratio cannot classify the quadrants on its own. Two negative loads give a positive ratio, indistinguishable from two positive ones. The sign pair is the primitive; the ratio describes position *within* a quadrant.

5.2 Two axes: quadrant and tier

The quadrant selects which processes are needed. The load *magnitude* selects which components can deliver them. The resulting mode set is:

Mode	Quadrant	Active components	Water required
M1	cooling + dehumidification	DW, cooler	$\geq 66^\circ\text{C}$
M2	cooling + dehumidification	DW, cooler, IEC	$\geq 66^\circ\text{C}$
M3	cooling + dehumidification	DW, cooler, ABS + CHW	$\geq 79^\circ\text{C}$
M4	cooling + humidification	humidifier alone (adiabatic saturation <i>is</i> direct evaporative cooling)	none
M5	cooling + humidification	ABS + CHW, humidifier	$\geq 79^\circ\text{C}$
M6	heating + dehumidification	DW, cooler and winter coil, <i>blended</i>	$\geq 66^\circ\text{C}$
M7	heating + humidification	winter coil	$\geq \text{supply } T$

Mode	Quadrant	Active components	Water required
M8	heating + humidification	winter coil, humidifier	$\geq$ supply T

Three of these carry conditions worth stating explicitly.

M6 is the only mode requiring modulating dampers. Dehumidification and heating sit on parallel prongs of fork 1, so the quadrant is reached by drying part of the flow, heating the rest, and blending at T_{pmo} . Every other mode is satisfied by two-position devices.

M5 has an unresolved ordering problem. The humidifier sits on the d_{pE} prong, upstream of the chilled-water coil in fork 2, so the air is humidified before it is cooled and may reach saturation. Cooling and then humidifying is the correct sequence, and the architecture does not currently provide it.

M7 and M8 route through prong D with the wheel stopped. Air passes a stationary wheel at nothing but a pressure drop, which turns the cooler into an exhaust-to-supply recuperator at no capital cost — see Section 6.3.

6 The resource constraint

6.1 Temperature: which tiers the well can serve

Only M3 and M5 require the 79°C minimum at the absorption generator. Every other mode needs approximately 66°C for wheel regeneration, which the well holds for the whole design life at almost any flow. The serviceable region of the load plane therefore *shrinks with reservoir depletion, and it shrinks from the cooling corner first*.

From the archived parametric sweep, year-20 conditions:

L [m]	BHT [°C]	q [m ³ /h]	T_{ret} [°C]	T_{wa} [°C]	Q_{well} [kW]	generator viable
4000	144.0	3	79.8	71.4	130	no
4000	144.0	4	78.7	72.2	168	no
4000	144.0	5	76.4	71.4	197	no
4000	144.0	6	73.8	69.8	220	no
4000	144.0	8	69.1	66.3	251	no
4000	144.0	10	65.4	63.4	270	no
4500	159.5	3	86.5	77.0	149	no
4500	159.5	4	86.4	78.9	198	yes
4500	159.5	5	84.5	78.5	237	yes
4500	159.5	6	82.0	77.2	267	no
4500	159.5	8	77.0	73.7	309	no
4500	159.5	10	72.8	70.4	336	no
4677	165.0	3	88.8	78.8	156	yes
4677	165.0	4	89.1	81.2	208	yes
4677	165.0	5	87.3	81.1	251	yes
4677	165.0	6	84.9	79.8	283	yes
4677	165.0	8	79.8	76.3	331	no
4677	165.0	10	75.6	73.0	360	no

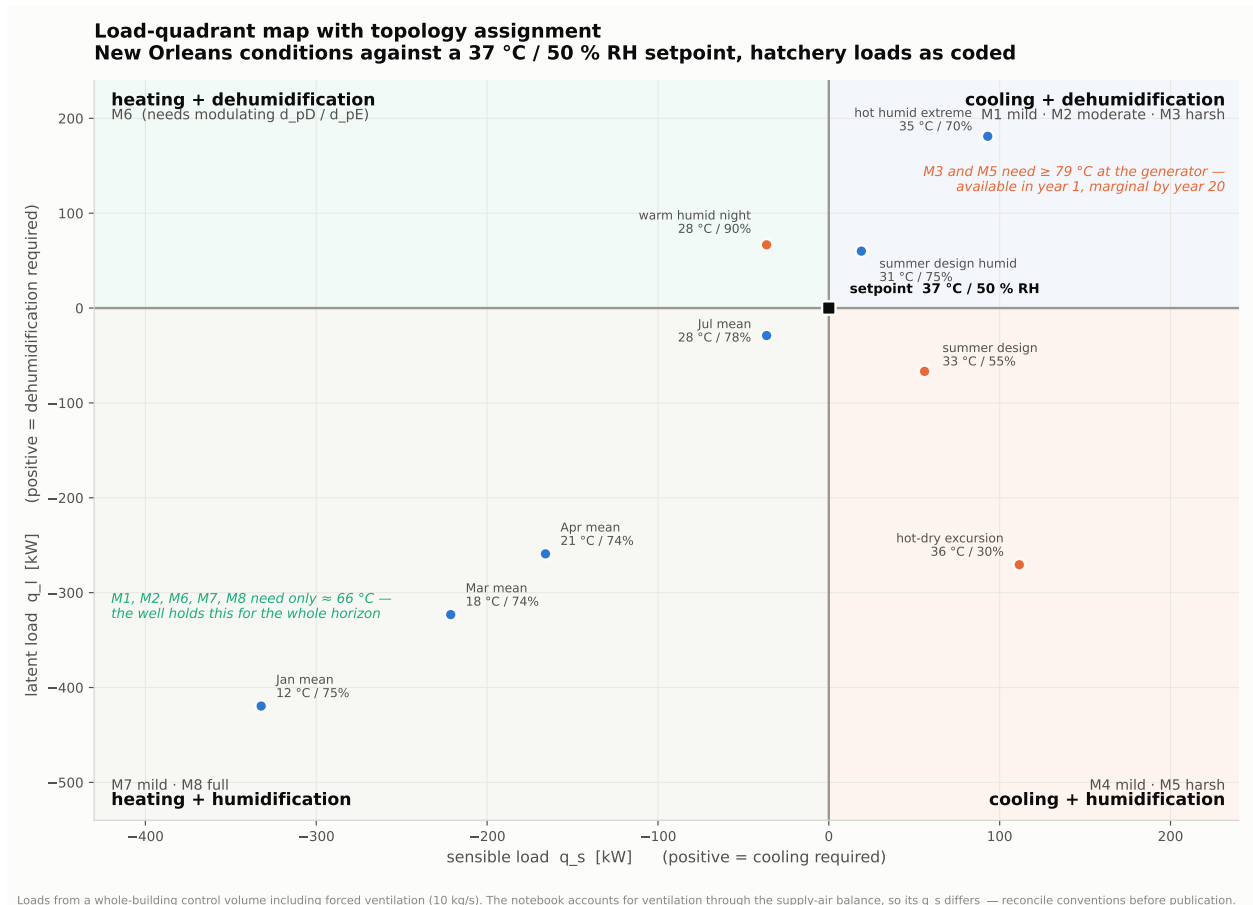


Figure 3: The load plane. Quadrants are named by the processes they require; the tier within a quadrant is set by load magnitude against what the well can deliver.

Topology variants — one per load quadrant and intensity tier

quadrants named by the processes they require; the tier says which components can deliver them

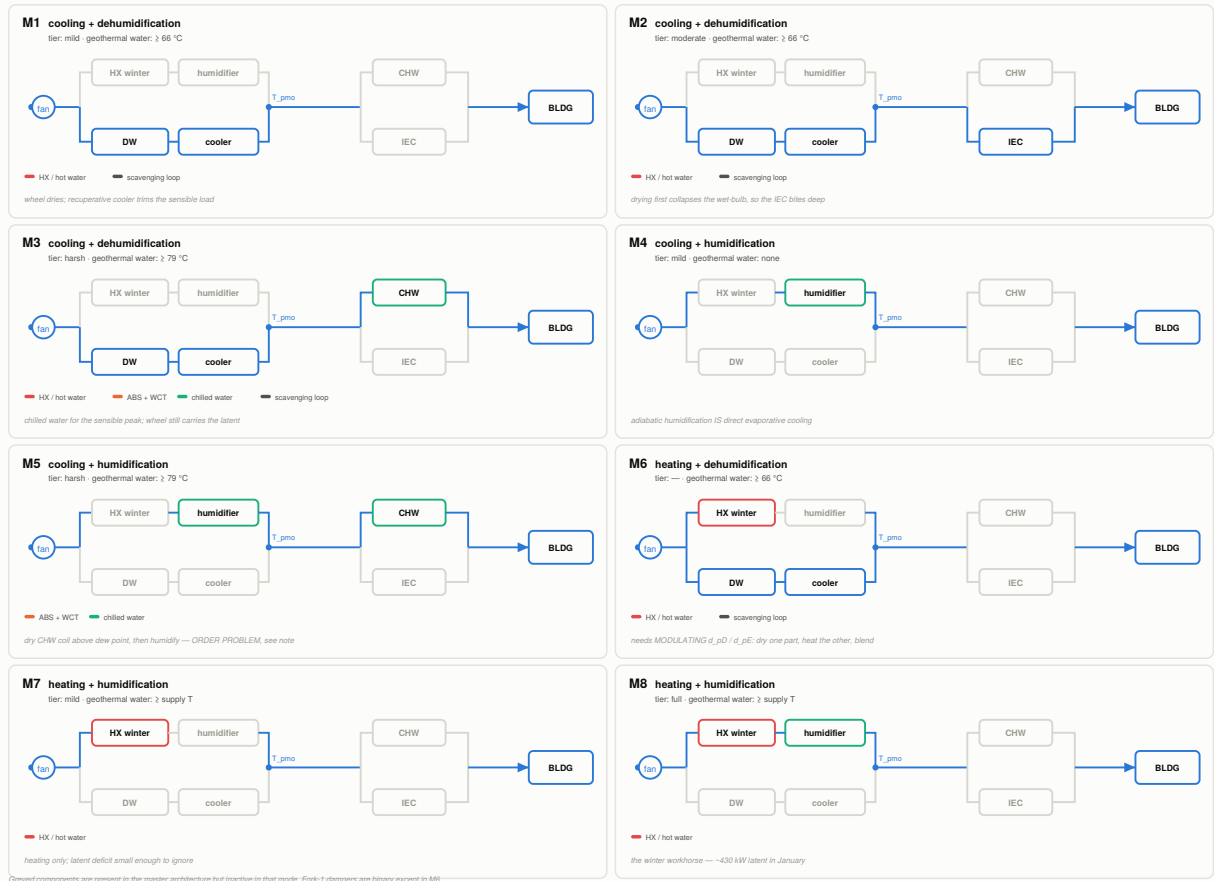


Figure 4: Topology variants. Each card shows the branches active in one mode; inactive branches remain installed but idle.

6.2 Capacity: the flow squeeze

Raising loop flow raises the generator duty but lowers the water temperature, and the two requirements are not satisfied at the same operating point. Using a design generator drop of 10 K and a maximum of 18 K, the coded sensible demand of 101 kW needs a generator duty of about 145 kW, that is roughly 7 m³/h — but at 4500 m that flow does not reach 79 °C. The temperature-optimal flow starves the chiller; the capacity-optimal flow cools the water below its minimum. Satisfying both requires depth: approximately 5000 m to 5500 m at 6 m³/h to 7 m³/h.

6.3 Winter capacity, and why recovery dominates

Sizing the January duty over the whole outdoor-air stream, the enthalpy lift from a cold, dry ambient to the comfort state is roughly 60 kJ/kg, giving a total near 772 kW of which more than half is latent. Net of internal gains the plant duty is about 630 kW, against a well output between 198 kW and 336 kW: the well covers only 31–53%.

A dry recuperator at 70% effectiveness recovers about 227 kW, leaving 403 kW unserved. A *latent-capable* device at the same effectiveness recovers about 540 kW, leaving 90 kW — comfortably inside the well’s capability. Because 58% of the winter duty is latent, only the second closes the gap.

This is the strongest argument in the architecture for routing M7 and M8 through prong D with the wheel halted: the cooler then acts as an exhaust-to-supply recuperator using hardware already installed. Whether the desiccant wheel itself, operated differently, can recover the latent half — a rotary wheel moving moisture from a warm humid exhaust to a cold dry supply *is* an enthalpy wheel — remains open.

7 Implementation status

Component	Status	Note
DBHE (GITT + coaxial BVP)	implemented	transient formation coupling, 20-year march
Thermal accumulator	implemented	lumped capacitance
Regeneration coil (HX)	implemented	effectiveness–NTU, water to air
Desiccant wheel	implemented	Jurinak F_1/F_2 with prescribed effectiveness
Recuperative cooler	implemented	process air to scavenging air
Load interface	implemented	v10.6; generic $(q_s, \dot{m}_w)$ with explicit closure
Humidifier	pending	adiabatic saturation; unlocks M4
Indirect evaporative cooler	pending	effectiveness form; unlocks M2
Winter heating coil	pending	reuses the coil model on different streams; unlocks M6–M8
Absorption chiller + tower + CHW	pending	unlocks M3, M5

A known limitation of the wheel model. The Jurinak formulation carries no size: no diameter, no desiccant mass, no rotation speed, no transfer area. The outlet is placed a fixed fraction of the way from the process inlet toward the regeneration inlet in F_1/F_2 space, and those fractions are the two prescribed effectivenesses. It can be verified directly that the process outlet state is *independent of both mass flows*. Three consequences follow: the wheel cannot be resized within this model, it

cannot be throttled by scavenging flow, and any claim about wheel sizing, rotation speed or part-load behaviour is outside what the formulation can represent. Deriving the effectivenesses from an NTU using the analogy method would remove these restrictions at essentially no computational cost, and is deferred.

8 Verification protocol

Each model version is frozen as a regression fixture recording every tracked state at annual resolution. Fixtures are named by version and are *never overwritten*: a mutable reference would always pass and prove nothing. A refactor that should not change physics is verified by reproducing the previous fixture to 1×10^{-9} ; a change that should move results is verified by explaining every value that moved.

The harness loads the notebook’s own code rather than re-implementing it, so if the two ever disagree the harness is wrong by definition. It is validated in both directions: an untouched notebook must pass, and a notebook perturbed by 0.45% in formation conductivity must fail.

A recurring failure mode. Four defects found so far shared one shape — a guard that degraded quietly instead of complaining:

1. a missing property library substituted constant water properties, misstating viscosity by up to 90% at the hot end;
2. a global warning filter suppressed the notebook’s own diagnostics, so a formation-model safety check could never fire;
3. the desiccant wheel could reverse sign and humidify the process air while the dashboard still reported the humidity target as met;
4. the scavenging mixing node conserved energy but not moisture, using two different damper positions for the two balances.

None raised an error. Each produced plausible output. The protocol now requires that any guard whose failure would otherwise be invisible must announce itself.

9 Version history

Year-20 values at the default operating point (4500 m, 10 m³/h, insulated tubing), taken directly from the frozen fixtures:

v	T_{ret} [°C]	T_{wa} [°C]	T_{ro} [°C]	w_{ri} [g/kg]	Δw [g/kg]	UA_{cool} [kW/K]	d_{rA} [—]
10.2	72.83	70.37	50.83	30.86	0.618	66.9	0.230
10.3	72.83	70.37	50.83	30.86	0.618	66.9	0.230
10.4	76.74	74.05	58.18	38.58	0.634	90.7	0.050
10.5	70.60	68.27	46.64	21.29	1.853	70.4	0.050
10.6	70.60	68.27	46.64	21.29	1.853	70.4	0.050

v10.2 Wheel-regime diagnostics; warning suppression fixed. Announced the desiccant-wheel sign reversal, and stopped `warnings.filterwarnings(ignore)` from swallowing the notebook’s own diagnostics.

v10.3 Generic load interface. The plant sees the building only through a two-equation, three-unknown interface with an explicit closure. Verified bit-identical to v10.2.

v10.4 Scavenging mixing-node mass balance. The moisture balance used the global damper constant while the enthalpy balance used the controlled one; the weights summed to 1.07 and moisture was not conserved.

v10.5 Scavenging fed from building exhaust. The loop had been recirculating the wheel's own desorbed moisture. Moisture removal nearly tripled.

v10.6 Two-sided humidity limit flag. The saturation flag could only see under-drying. Reporting only; verified bit-identical to v10.5.

On the two largest movements. The mass-balance correction at v10.4 moved results in the direction *opposite* to the one predicted. Fixing the balance improves the wheel at any given damper position, so the controller needs less venting and drives the damper to its floor; at high recirculation the loop accumulates more of the wheel's own moisture, and the regeneration inlet ends up wetter rather than drier. The open-loop estimate and the closed-loop result disagreed in sign. At v10.5, removing recirculation entirely then reduced the regeneration inlet humidity by roughly 17 g/kg and nearly tripled moisture removal.

Current open item. The humidity controller is saturated at its dry end: the supply is drier than the space requires and the damper has no authority left to reduce drying. Approximately 41% of the regeneration coil duty is presently spent removing moisture the building does not want. The remedy is to move humidity control from the scavenging damper to v_G , the share of water flow reaching the regeneration coil, which throttles the coil duty directly and returns that heat to the loop.